



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to N.J.K FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

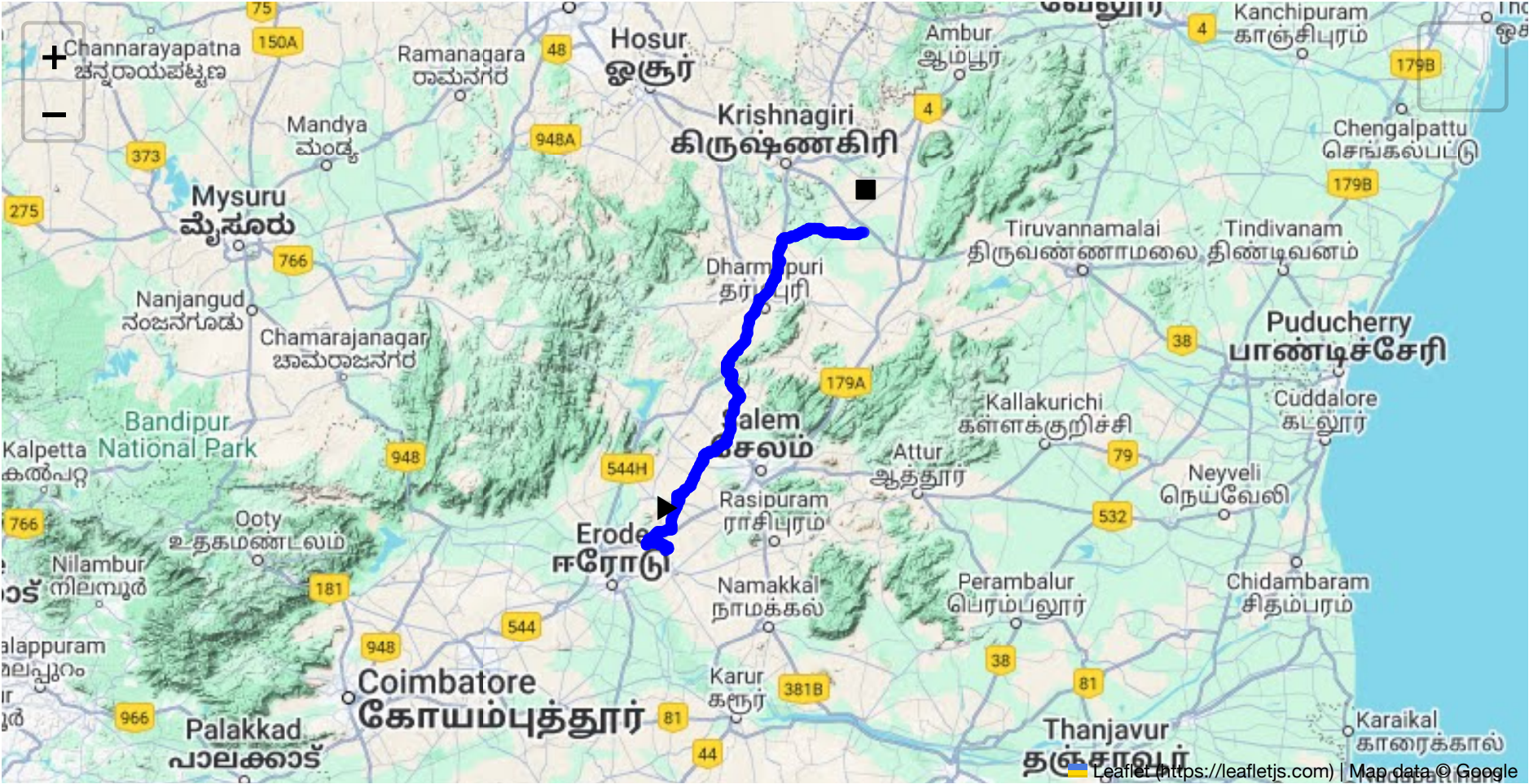
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 154.89 km
Estimated Duration: 3.2 hours
Adjusted Duration (Heavy Vehicle): 4.0 hours

Start: (11.4381, 77.8734)
End: (12.333988, 78.440232)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 154.89 kilometers through Tamil Nadu, India, starting from CVQF+23W in Sangagiri and ending at 8CMR+G3W on NH 77 in Marappanayakkampatti. The journey includes key waypoints at Bhavani Main Road in Padaiveedu, SH 86 in Ayveli, and Olakkachinnanur. It primarily traverses state highways and local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region typically experiences a tropical climate with hot summers, mild winters, and a monsoon season from June to September. Potential hazards include:

- Heavy rainfall during monsoon causing road flooding and reduced visibility.
- High temperatures in summer leading to road surface damage.
- Occasional fog in winter, especially in early mornings, affecting visibility.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (7-9 AM) and evening (5-8 PM) when local commuters and transport vehicles are most active.
- **Congestion-Prone Areas:** Urban stretches near major towns and intersections, especially around Erode and Salem districts.

4. Assessment of Road Quality and Infrastructure

- The road quality varies, with well-maintained stretches on state highways but potential potholes and narrow sections in rural and less urbanized areas.
- Key infrastructure includes road signs, occasional narrow bridges, and level crossings that may impact heavy vehicle navigation.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** If NH 77 becomes inaccessible, consider using alternate state highways connecting back to major roads leading to the destination.
- Keep updated with local traffic advisories for real-time alternative suggestions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Adhere to state regulations requiring permits for hazardous materials.
- Restrictions may exist on travel during night times or through certain populated areas.
- Follow speed limits and signage especially concerning heavy vehicles and hazardous loads.

7. Overview of Historical Incidents

- Few documented incidents involving heavy vehicles; however, typical accidents involve overturned trucks due to road conditions and driver errors.
- Adherence to regulations and caution can mitigate risks.

8. Environmental Considerations and Sensitive Areas

- Several water bodies and agricultural fields, where spillages could cause significant harm.
- Respect local wildlife areas and agricultural zones to prevent environmental damage.

9. Analysis of Communication Coverage

- Mobile network is generally reliable along major highways, but potential dead zones may exist in rural stretches.
- Ensure redundancy in communication methods for safety.

10. Estimated Emergency Response Times

- **Urban Areas:** 20-30 minutes due to proximity to facilities.
- **Rural Stretches:** Can extend to 60 minutes or more due to less accessible roads.

11. Overall Summary of Risk Assessment

The route features a mix of well-maintained highways and challenging rural roads. Weather conditions and road quality are the primary concerns, along with localized congestion near major towns. Proper adherence to safety regulations and preparedness for emergency diversions can mitigate most risks. It is recommended to conduct a pre-trip inspection and continuously monitor the conditions throughout the journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	Medium	11.45546, 77.81800	30 KM/Hr
7	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
8	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
9	Turn	High	11.49469, 77.88488	15 KM/Hr
10	Turn	High	11.55614, 77.89764	15 KM/Hr
11	Turn	Medium	11.55680, 77.89969	30 KM/Hr
12	Turn	High	11.58218, 77.90845	15 KM/Hr
13	Turn	High	11.58330, 77.90642	15 KM/Hr
14	Turn	Medium	11.59189, 77.90645	30 KM/Hr
15	Turn	Medium	11.59328, 77.90809	30 KM/Hr
16	Turn	Medium	11.63867, 77.94841	30 KM/Hr
17	Turn	High	11.64217, 77.95200	15 KM/Hr
18	Turn	Medium	11.64712, 77.95118	30 KM/Hr
19	Turn	Medium	11.67858, 77.96834	30 KM/Hr
20	Turn	Medium	11.70062, 77.97961	30 KM/Hr
21	Turn	High	11.70218, 77.97866	15 KM/Hr
22	Turn	Medium	11.71399, 77.99580	30 KM/Hr
23	Turn	Medium	11.71674, 77.99901	30 KM/Hr
24	Turn	Medium	11.71917, 78.01123	30 KM/Hr
25	Turn	Medium	11.73769, 78.03451	30 KM/Hr
26	Turn	High	11.73651, 78.03871	15 KM/Hr
27	Turn	Medium	11.74143, 78.04613	30 KM/Hr
28	Turn	Medium	11.86847, 78.08367	30 KM/Hr
29	Turn	Medium	11.87942, 78.08197	30 KM/Hr
30	Turn	Medium	11.92585, 78.06507	30 KM/Hr
31	Turn	Medium	11.93122, 78.06489	30 KM/Hr
32	Turn	Medium	11.93447, 78.06305	30 KM/Hr
33	Turn	Medium	11.93704, 78.05402	30 KM/Hr
34	Turn	Medium	11.95664, 78.04781	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
35	Turn	High	11.95818, 78.04978	15 KM/Hr
36	Turn	Medium	11.95988, 78.04948	30 KM/Hr
37	Turn	Medium	11.96304, 78.05136	30 KM/Hr
38	Turn	High	11.96399, 78.05135	15 KM/Hr
39	Turn	High	11.96580, 78.04923	15 KM/Hr
40	Turn	Medium	11.96675, 78.04939	30 KM/Hr
41	Turn	Medium	11.96791, 78.05070	30 KM/Hr
42	Turn	Medium	12.16983, 78.16769	30 KM/Hr
43	Turn	Medium	12.19881, 78.18157	30 KM/Hr
44	Turn	Medium	12.22459, 78.19136	30 KM/Hr
45	Turn	Medium	12.24731, 78.19601	30 KM/Hr
46	Turn	Medium	12.25060, 78.19977	30 KM/Hr
47	Turn	Medium	12.25329, 78.20039	30 KM/Hr
48	Turn	Medium	12.29110, 78.19909	30 KM/Hr
49	Turn	Medium	12.31483, 78.20789	30 KM/Hr
50	Turn	High	12.31904, 78.21608	15 KM/Hr
51	Turn	Medium	12.31827, 78.21725	30 KM/Hr
52	Turn	Medium	12.32435, 78.24037	30 KM/Hr
53	Turn	High	12.32373, 78.24193	15 KM/Hr
54	Turn	Medium	12.33753, 78.27092	30 KM/Hr
55	Turn	Medium	12.34226, 78.28138	30 KM/Hr
56	Turn	Medium	12.34566, 78.28283	30 KM/Hr
57	Turn	Medium	12.34341, 78.30518	30 KM/Hr
58	Turn	Medium	12.33200, 78.35662	30 KM/Hr
59	Turn	Medium	12.33289, 78.35872	30 KM/Hr
60	Turn	High	12.33199, 78.36425	15 KM/Hr
61	Turn	High	12.33652, 78.37455	15 KM/Hr
62	Turn	High	12.32934, 78.40089	15 KM/Hr
63	Turn	High	12.33043, 78.40180	15 KM/Hr
64	Turn	High	12.32927, 78.41871	15 KM/Hr
65	Turn	High	12.33099, 78.42022	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
12	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
15	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
16	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
17	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
18	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
24	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	fire_station	Fire Station	12.3328308, 78.3489464	30 km/h	Medium
30	hospital	Government Hospital, Pochampalli	12.3322477, 78.365767	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
27	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
28	school	Chellampatti -PUPS	12.3467776, 78.2947171	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55614, 77.89764



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58330, 77.90642



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59328, 77.90809



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63867, 77.94841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64217, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64712, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70062, 77.97961



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70218, 77.97866



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71399, 77.99580



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71674, 77.99901



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71917, 78.01123



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74143, 78.04613



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93122, 78.06489



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93447, 78.06305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95664, 78.04781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.16983, 78.16769



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



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Coordinates: 12.22459, 78.19136



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Speed Limit: 30 KM/Hr
Coordinates: 12.24731, 78.19601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25060, 78.19977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31483, 78.20789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31904, 78.21608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31827, 78.21725



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32435, 78.24037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.32373, 78.24193



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.34341, 78.30518



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.33200, 78.35662



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.33289, 78.35872



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.33199, 78.36425



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.33652, 78.37455
